Latihan 2 Matematika Terapan

No.

Date:

Nama : Devita Sylfa Azzahraa

Nim : 2213310009

Kelas : 1A TTE

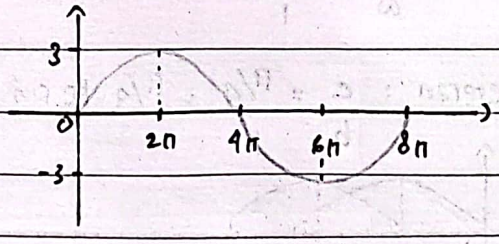
Bagian 1

Tentukan amplitudo, periode dan gambarkan grafik fungsi trigonometri berikut untuk 1 T!

1. $y = 3 \sin \frac{1}{4} x$

Amplitudo : 3

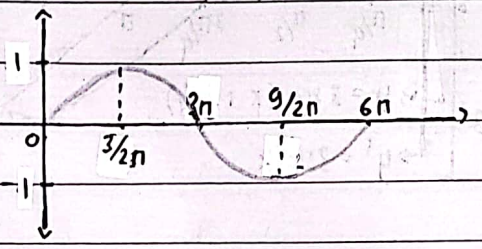
Periode : $\frac{2\pi}{\omega} = \frac{2\pi}{1/4} = 8\pi$



2. $y = \sin \frac{1}{3} x$

Amplitudo : 1

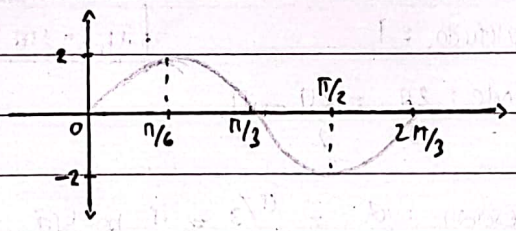
Periode : $\frac{2\pi}{\omega} = \frac{2\pi}{1/3} = 6\pi$



3. $y = 2 \sin 3x$

Amplitudo : 2

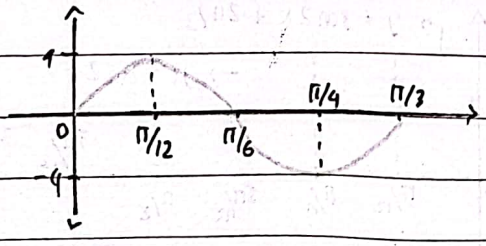
Periode : $\frac{2\pi}{\omega} = \frac{2\pi}{3}$



4. $y = 4 \sin 6x$

Amplitudo : 4

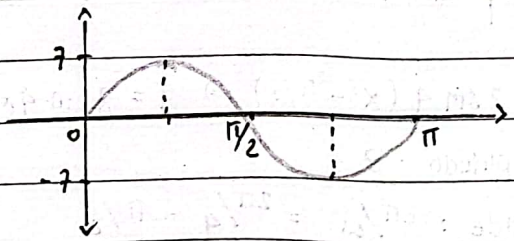
Periode : $\frac{2\pi}{\omega} = \frac{2\pi}{6} = \frac{\pi}{3}$



7. $y = 7 \cos 2t$

Amplitudo : 7

Periode : $\frac{2\pi}{\omega} = \frac{2\pi}{2} = \pi$



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Bagian II

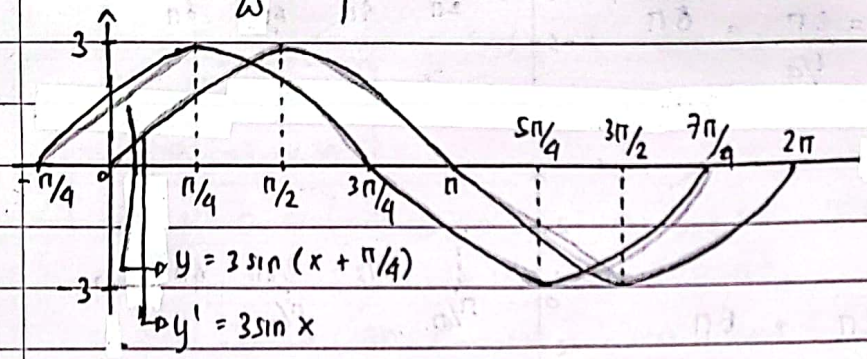
Tentukan amplitudo, periode dan sudut pergeseran kemudian gambarkan grafik fungsi trigonometri untuk satu periode!

1. $y = 3\sin(x + \pi/4)$

Amplitudo : 3 $\quad y_1 = 3\sin x$

Periode : $\frac{2\pi}{\omega} = \frac{2\pi}{1} = 2\pi$

Pergeseran : $\frac{a}{\omega} = \frac{\pi/4}{1} = \pi/4$ ke kiri

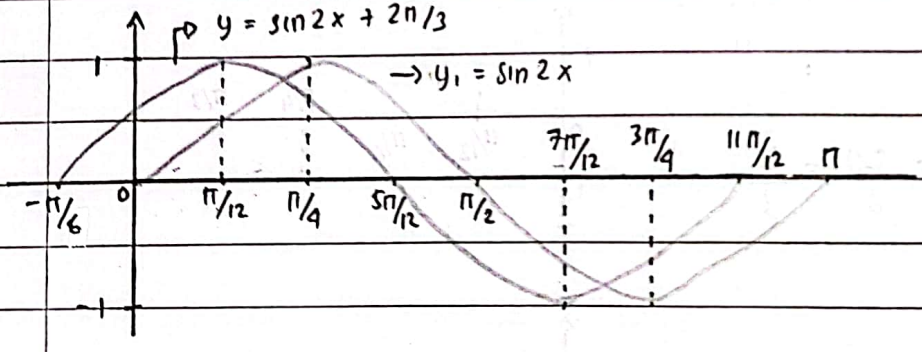


3. $y = \sin 2(x + \pi/3) \Rightarrow y = \sin 2x + 2\pi/3$

Amplitudo : 1 $\quad y_1 = \sin 2x$

Periode : $\frac{2\pi}{\omega} = \frac{2\pi}{2} = \pi$

Pergeseran : $\frac{a}{\omega} = \frac{\pi/3}{2} = \frac{\pi}{6}$ ke kiri



2. $y = 3\sin 4(x - \pi/3) \Rightarrow y = 3\sin 4x - 4\pi/3$ $\quad y_1 = 3\sin 4x$

Amplitudo : 3

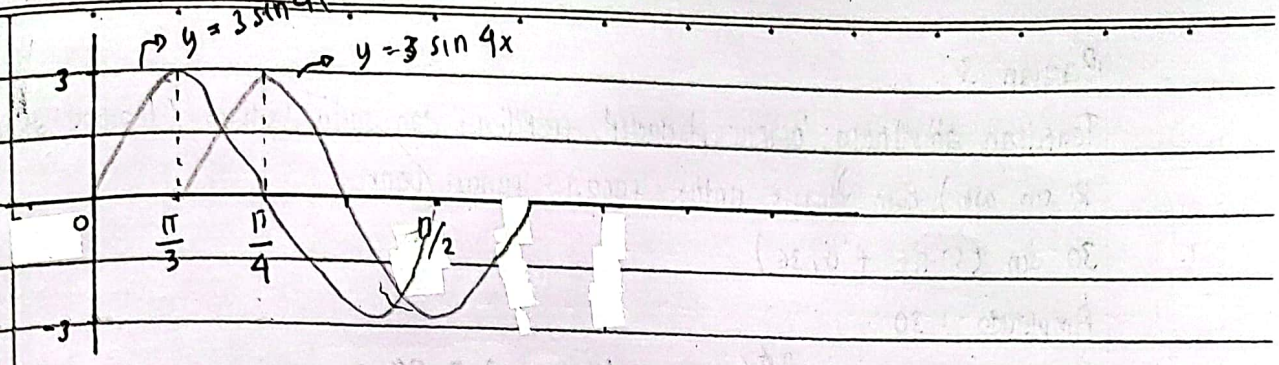
Periode : $2\pi/\omega = 2\pi/4 = \pi/2$

Pergeseran : $\frac{a}{\omega} = \frac{-4\pi/3}{4} = \frac{\pi}{3}$ ke kanan

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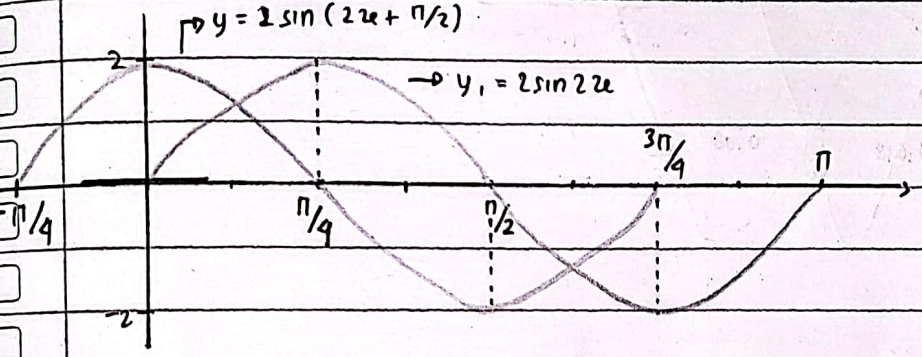


4. $y = 2 \sin (2x + \pi/2)$ $\quad y_1 = 2 \sin 2x$

Amplitudo : 2

Periode : $2\pi/\omega = 2\pi/2 = \pi$

Pergeseran : $\frac{a}{\omega} = \frac{\pi/2}{2} = \pi/4$ ke kiri

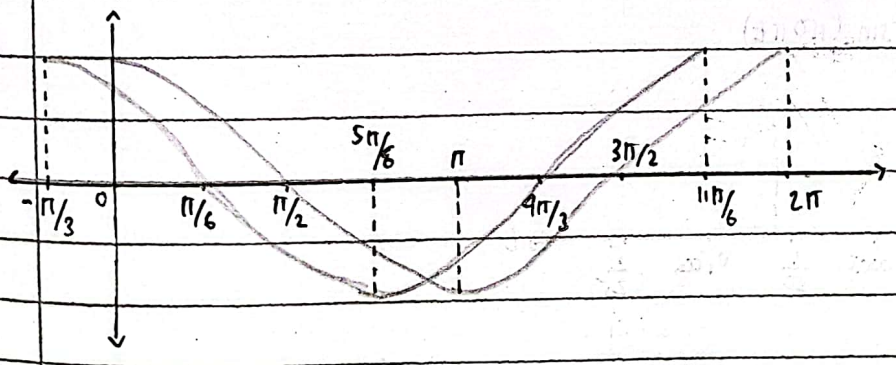


6. $y = 2 \cos (x + \pi/3)$ $\quad y_1 = 2 \cos x$

Amplitudo : 2

Periode : $2\pi/\omega = 2\pi/1 = 2\pi$

Pergeseran : $\frac{a}{\omega} = \frac{\pi/3}{1} = \frac{\pi}{3}$ ke kiri



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Bagian 3

Tentukan amplitudo, waktu periodik, frekuensi dan sudut phase (leading atau legging R sin ωt) dan grafik untuk fungsi - fungsi berikut :

1. $30 \sin (50\pi t + 0{,}36)$

Amplitudo : 30

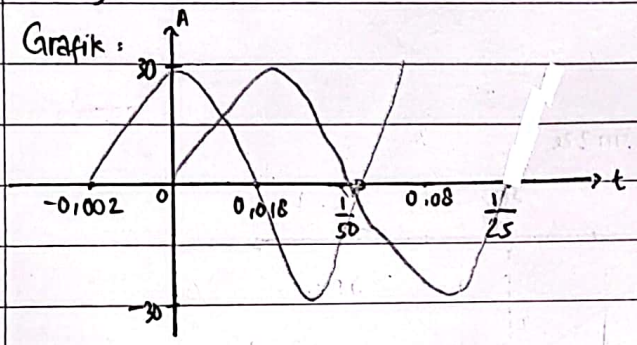
Periode : $2\pi/\omega = 2\pi/50\pi t = 1/25$ s.

Sudut phase $\frac{\alpha}{\omega} = \frac{-0{,}36 \text{ rad}}{50\pi t} = 0{,}002$ s ke kiri

Frekuensi : $\frac{1}{T} = \frac{1}{\frac{1}{25}s} = 25$ Hz

Leading : $30 \sin (50\pi t)$

Grafik :



5. $y = 63{,}5 \sin (150\pi t - 0{,}132)$

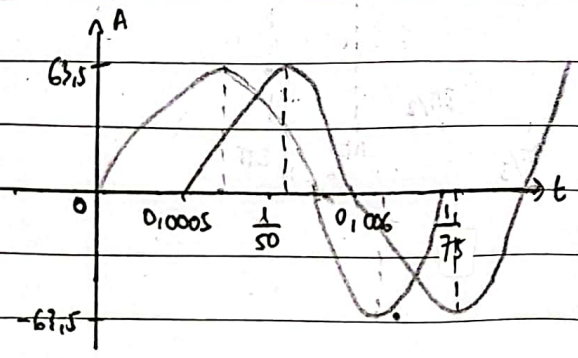
Amplitudo : 63,5

Periode : $2\pi/150\pi t = 1/75$ detik

Sudut phase : $\frac{\alpha}{\omega} = \frac{0{,}132 \text{ rad}}{150\pi t} = 0{,}0005$ sec ke kanan

Frekuensi : $1/T = 1/\frac{1}{75}s = 75$ Hz

Leading : $63{,}5 \sin (150\pi t)$



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Analisis Bagian 3

No.

Date:

Nama : Devita Syifa Azzahraa

Nim : 2213310009

Kelas : 1A - TTE

1. $y = 30 \sin (50\pi t + 0{,}36)$

a. y bila t = 0

$y = 30 \sin (0{,}36)$

$= 30 (0{,}3522)$

$= 10{,}56$

b. y bila t = 2 ms → 0,002 s

$y = 30 \sin (50 . 3{,}14 . 0{,}002 + 0{,}36)$

$= 30 \sin (0{,}314 + 0{,}36)$

$= 30 \sin (0{,}674)$

$= 30 . 0{,}6241$

$= 18{,}723$

c. t bila y = 15

$y = 30 \sin (50\pi t + 0{,}36)$

$15 = 30 \sin (50\pi t + 0{,}36)$

$15/30 = \sin (50t + 0{,}36)$

arc sin 0,5 = $50t + 0{,}36$

$\pi/6 = 50\pi t + 0{,}36$

$t = \dfrac{\pi/6 - 0{,}36}{50\pi}$

sin yg hasilnya 0,5

Cari t_1

$t_1 = \dfrac{\pi/6 - 0{,}36}{50\pi}$

$= \dfrac{3{,}14/6 - 0{,}36}{50(3{,}14)}$

$= \dfrac{0{,}523 - 0{,}36}{157}$

$= \dfrac{0{,}163}{157}$

$= 0{,}001$ sekon

Cari t_2 → $(180 - \alpha) = 180 - 30 = 150$ → ubah ke rad

$150 \cdot \dfrac{\pi}{180} = \dfrac{5\pi}{6}$

$t_2 = \dfrac{5\pi/6 - 0{,}36}{50\pi}$

$= \dfrac{5 . 3{,}14/6 - 0{,}36}{50(3{,}14)}$

$= \dfrac{2{,}616 - 0{,}36}{157}$

$= 0{,}014$ sekon

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d. t_1 & t_2 bila y = maks → 30

$y = 30 \sin (50\pi t + 0{,}36)$

$30 = 30 \sin (50\pi t + 0{,}36)$

$30/30 = \sin (50\pi t + 0{,}36)$

→ $\arcsin 1 = 50\pi t + 0{,}36$

$\pi/2 = 50\pi t + 0{,}36$

$$t = \frac{\pi/2 - 0{,}36}{50\pi}$$

$$t_1 = \frac{\pi/2 - 0{,}36}{50\pi} = \frac{3{,}14/2 - 0{,}36}{50(3{,}14)} = \frac{1{,}57 - 0{,}36}{157} = 0{,}007 \text{ sekon}$$

$$t_2 = \frac{5\pi/2 - 0{,}36}{50\pi} = \frac{5 \times 3{,}14/2 - 0{,}36}{50(3{,}14)} = \frac{7{,}49}{157} = 0{,}04477 \text{ sekon}$$

e. t_1 dan t_2 bila $y = 0$

$0 = 30 \sin (50\pi t + 0{,}36)$

$0/30 = \sin (50\pi t + 0{,}36)$

$\arcsin 0 = 50\pi t + 0{,}36$

$\pi = 50\pi t + 0{,}36$

$$t = \frac{\pi - 0{,}36}{50\pi}$$

→ $t_1 = \frac{\pi - 0{,}36}{50\pi} = \frac{3{,}14 - 0{,}36}{157} = 0{,}017$ sekon

$t_2 = \frac{2\pi - 0{,}36}{50\pi} = \frac{2 \cdot 3{,}14 - 0{,}36}{50(3{,}14)}$

→ $t_2 = \frac{6{,}28 - 0{,}36}{157} = 0{,}037$ sekon

$(180 + \alpha)$
$= 180 + 180$
$= 360$ → ubah rad
$= 2\pi$

5. $y = 63{,}5 \sin (150\pi t - 0{,}132)$

a. y bila $t = 0$

$y = 63{,}5 \sin (-0{,}132)$

$y = 63{,}5 (-0{,}1316)$

$= -8{,}3566$

b. y bila t = 2 ms (0,002 sekon)

$y = 63{,}5 \sin (150\pi t - 0{,}132)$

$= 63{,}5 \sin (150 \cdot 3{,}14 \cdot 0{,}002 - 0{,}132)$

$= 63{,}5 \sin (0{,}942 - 0{,}132)$

$= 63{,}5 \sin (0{,}81)$

$= 63{,}5 \times 0{,}7292$

→ 45,986

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c. t bila $y = 15$

$y = 63{,}5 \sin(150\pi t - 0{,}132)$

$15 = 63{,}5 \sin(150\pi t - 0{,}132)$

$15/63{,}5 = \sin(150\pi t - 0{,}132)$

$0{,}236 = \sin(150\pi t - 0{,}132)$

$\arcsin 0{,}236 = 150\pi t - 0{,}132$

$0{,}238 = 150\pi t - 0{,}132$

$t_1 = \frac{0{,}238 + 0{,}132}{150\pi} = \frac{0{,}37}{471} = 0{,}00078$ sekon

$t_2 = t_1 + T$

$= 0{,}00078 + 0{,}0133$

$= 0{,}0141$ sekon

d. t_1 dan t_2 bila y = maks (30)

$y = 63{,}5 \sin(150\pi t - 0{,}132)$

$30 = 63{,}5 \sin(150\pi t - 0{,}132)$

$30/63{,}5 = \sin(150\pi t - 0{,}132)$

$0{,}4172 = \sin(150\pi t - 0{,}132)$

$\arcsin 0{,}472 = 150\pi t - 0{,}132$

$0{,}4915 = 150\pi t - 0{,}132$

$t_1 = \frac{0{,}4915 + 0{,}132}{150\pi} = \frac{0{,}6235}{471} = 0{,}00132$ sekon

$t_2 = t_1 + T$

$= 0{,}00132 + 0{,}0133$

$= 0{,}01462$ sekon

e. t_1 dan t_2 bila $y = 0$

$y = 63{,}5 \sin(150\pi t - 0{,}132)$

$0 = 63{,}5 \sin(150\pi t - 0{,}132)$

$0/63{,}5 = \sin(150\pi t - 0{,}132)$

$0 = \sin(150\pi t - 0{,}132)$

$\arcsin 0 = 150\pi t - 0{,}132$

$\pi = 150\pi t - 0{,}132$

$t_1 = \frac{\pi + 0{,}132}{150\pi} = \frac{3{,}14 + 0{,}132}{471} = 0{,}0069$ sekon

$t_2 = t_1 + T$

$= 0{,}0069 + 0{,}0133$

$= 0{,}1399$ sekon

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